

Foundations of Supervised and Unsupervised Machine Learning

Course Syllabus

Format: 12 lectures, 1 session/week, 1h45 each

References:

- ISL / James, Witten, Hastie, Tibshirani & Taylor — *An Introduction to Statistical Learning* (2nd ed., 2023)
- ESL / Hastie, Tibshirani & Friedman — *The Elements of Statistical Learning* (2nd ed., 2009)
- DL / Bishop & Bishop — *Deep Learning: Foundations and Concepts* (2024)
- PRML / Bishop — *Pattern Recognition and Machine Learning* (2006)

Note: This is a very tentative plan. The order, content, and pacing of lectures may be substantially revised as the semester progresses.

Course Overview

Lecture 1 / Introduction. A review of the probability and statistics background used throughout the course.

Lectures 2–8 / Supervised Learning. Learning to predict an output from labeled examples. We start with the two most basic strategies for regression and classification, then progressively build up ways to control model complexity and fight overfitting—first through ensembles of trees, then through regularization and kernel methods—before formalizing these ideas with a rigorous treatment of the bias-variance tradeoff and model selection. The block closes with support vector machines as a further case study.

Lectures 9–12 / Unsupervised Learning. Learning structure from data without labels: clustering, dimensionality reduction, probabilistic graphical models, and latent-variable models such as mixture models, the EM algorithm, and hidden Markov models.

Lecture Plan

#	Block	Topic
1	—	Introduction; probability and Gaussian toolbox
2	Supervised	Statistical learning: overview & key-concepts
3	Supervised	Linear regression
4	Supervised	Classification
5	Supervised	Trees and random forests
6	Supervised	Addressing overfitting: regularization, kernels, basis functions

#	Block	Topic
7	Supervised	Bias-variance tradeoff and model selection
8	Supervised	Support vector machines
9	Unsupervised	Unsupervised learning; clustering
10	Unsupervised	Principal component analysis and related methods
11	Unsupervised	Graphical models
12	Unsupervised	Mixture models, EM, and hidden Markov models